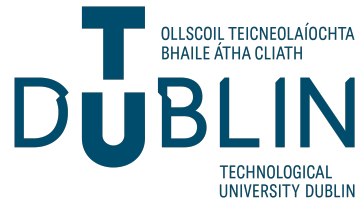


TU Dublin – Blanchardstown Campus
School of Informatics and Engineering



BN529: Master of Science in Applied Data Science and Analytics

M.Sc. Research Project

Title goes here

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August 31, 2026

Declaration

I hereby certify that this material, which I now submit for assessment on the programme of study leading to the award of MSc in Computing in the Institute of Technology Blanchardstown, is entirely my own work except where otherwise stated, and has not been submitted for assessment for an academic purpose at this or any other academic institution other than in partial fulfilment of the requirements of that stated above.

Author: Mariem Nsiri

Dated: August 31, 2026



Dedicated to . . .

Acknowledgment

Thanks to ...

Abstract

Abstract goes here. The purpose of the abstract is to provide a sufficient overview of the thesis so a reader can determine its relevance to their own research. It should cover: the context and aim of the study; key points of methodology; key results; and the significance/impact of those results. Don't include citations in the abstract, stick to the key points of this study.

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Chapter 1

Introduction

The rapid advancement of Generative Artificial Intelligence (GenAI) is reshaping the educational landscape, creating new possibilities for personalised learning, administrative automation, and pedagogical innovation. At the same time, it is introducing significant uncertainty into existing educational systems, requiring institutions, policymakers, and educators to reconsider how teaching, learning, and assessment are designed, governed, and evaluated. As AI tools become more accessible and increasingly embedded in everyday educational practice, the central challenge is no longer whether AI will affect education, but how this transformation can be governed in ways that are pedagogically meaningful, ethically responsible, and socially equitable.

International organisations have responded to this challenge by developing frameworks for responsible AI use in education. In particular, Miao and Cukurova [2024] and Miao et al. [2024], in UNESCO’s *AI Competency Frameworks for Teachers and Students*, emphasise that effective AI integration depends not only on the availability of technological tools, but on teacher capability, student AI literacy, institutional readiness, ethical awareness, and equity of access. This perspective reflects a broader shift in the literature, where AI in education is increasingly understood not as a technical issue, but as a *governance problem* requiring alignment between policy goals, implementation, and the realities of educational practice.

However, the literature consistently shows that such alignment remains difficult to achieve. Across policy and research studies, a recurring challenge concerns the translation of high-level principles (e.g., transparency, fairness, safety, and accountability) into practical and context-specific action within educational settings. In practice, educators are often required to interpret and enact policy in fast-changing

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environments without clear procedural guidance, robust training, or shared institutional norms. This creates a persistent gap between policy aspiration and educational enactment.

This policy–practice gap is also reflected in up-to-date studies. Ng et al. [2025] highlights that while AI offers substantial opportunities for adaptive learning, feedback generation, and administrative efficiency, these benefits are constrained by concerns regarding teacher preparedness, ethical use, academic integrity, and unequal access to digital resources. Similar patterns are evident in the systematic study by Bagdonaite and Dagiene [2025], which shows that enthusiasm for AI is often accompanied by uncertainty about its pedagogical value, its implications for student thinking, and the conditions under which its use can be considered legitimate and educationally sound. Taken together, these findings suggest that the problem is not simply one of innovation uptake alone, but of implementation capacity, institutional readiness, and governance coherence.

Research Problem
Despite the proliferation of AI governance frameworks in education, a persistent gap exists between policy design and classroom implementation. This gap is driven by limited operational guidance, insufficient teacher preparedness, and unequal access to resources, constraining the effective, ethical, and inclusive integration of AI in educational practice.

Evidence from policy and public sentiment further underlines this challenge. In Ireland, for example, recent government guidance by Department of Education and Youth [2025] acknowledges the need to support schools in managing both the opportunities and the risks of AI use, especially in relation to assessment, ethics, and adoption. At the same time, Quality and Qualifications Ireland [2025] and Kirk et al. [2023] suggest a more nuanced picture: stakeholders recognise AI’s potential, but concerns about trust, regulation, fairness, and practical guidance remain. Comparable patterns also appear internationally. In the United States, McClain et al. [2026] shows growing familiarity with AI, especially among younger users, but also concerns around trust, verification, and the social impact of AI-generated content. Together, these studies suggest a broader conclusion: successful AI adoption depends not only on policy design, but also on how such policy is interpreted and enacted by those responsible for its implementation.

Addressing this problem requires moving beyond descriptive accounts of AI policy toward an analysis of how policy is formulated, communicated, interpreted, and enacted. The policy–practice gap should be treated not simply as an administrative problem, but as a layered governance issue that emerges across formal policy documents, institutional responses, and stakeholder discourse. This study was therefore guided by the following research question:

Research Question
How can a multi-layered Natural Language Processing (NLP) framework be applied to identify, analyse, and explain the policy–practice gap in AI-in-education governance across different national contexts?

To answer the research question, this study adopted a computational “text-as-data” approach. This choice is grounded in the literature review, which shows that AI governance, public perceptions, and educational challenges are often examined separately, despite their close interconnections. The value of a text-as-data approach lies in its capacity to analyse policy documents and stakeholder discourse systematically, at a scale that would be difficult to achieve through manual reading alone. By combining topic modelling, sentiment analysis, and selected elements of causal inference, the study sought to identify *recurring themes*, detect *areas of tension or misalignment*, and examine *relationships* between policy language, stakeholder sentiment, and implementation challenges. In doing so, it moved beyond a descriptive account of AI policy and developed an integrated analytical framework for explaining how the policy–practice gap emerges, where it becomes visible, and how it is sustained across different governance settings.

Drawing on UNESCO’s AI Competency Frameworks by Miao and Cukurova [2024] and Miao et al. [2024], which emphasise human-centred, ethical, pedagogical, technical, and professional dimensions of AI competency, the analysis is organised around four dimensions: 1) policy and strategy, 2) implementation in practice, 3) teacher and student competency development, and 4) equity of access. Together, these dimensions help examine how AI-in-education policy moves from intention toward practice. They direct attention to the implementation conditions, including institutional capacity, teacher readiness, learner preparedness, access to resources, and how policy commitments become actionable in education.

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These dimensions are also related to the study of public sentiment. Questions of trust, fairness, teacher preparedness, access, and ethical use are not only present in policy frameworks, but also shape how AI in education is perceived by teachers, students, parents, and wider society. For this reason, the study treated public sentiment as an important layer of the policy–practice gap. It provides evidence of how policy priorities are received, questioned, or supported by those affected by AI adoption in educational settings.

The study adopted a comparative perspective across Ireland, France, Australia, and the United States. These countries were selected because they offer useful contrasts in governance structure, policy tradition, and educational response to AI. Ireland is included as the national context of the study and as a smaller education system operating within wider European regulatory developments. France provides a contrasting case because of its more centralised governance model and stronger emphasis on legal and regulatory structuring. The United States is included because of its global influence in AI development and its highly decentralised educational landscape, where implementation is shaped by variation across states, districts, and institutions. Australia offers an important intermediate case: according to Australian Government Department of Education [2023], it is a federal system that combines national coordination with local variation, and it was among the earlier national systems to develop dedicated guidance on the responsible use of GenAI in schools. Together, these cases support comparison across centralised, decentralised, and hybrid governance settings with different implementation capacities.

At the same time, the public sentiment component of the study was approached with a more focused scope. Because comprehensive and directly comparable public sentiment data is difficult to collect across all national contexts, the study gave particular attention to European evidence on public attitudes toward AI and GenAI in education. This evidence was used as a cautious proxy for broader Western trends, since Poushter et al. [2025] shows that European, North American, and Australian contexts often share concerns around trust, ethical use, regulation, and the social consequences of AI, while still allowing for national differences in how these concerns are expressed. To address gaps and imbalance in the available sentiment evidence, the study also used synthesised sentiment data. This draws on Imran et al. [2022] and Bayer et al. [2022], which show that synthetic text generation and data augmentation can support sentiment analysis and text classification in limited

or imbalanced data settings. This supports a more balanced comparison between policy discourse and public sentiment, provided that systematic validation is used to assess the faithfulness of the generated data to the original evidence, including its semantic consistency, thematic alignment, and divergence from the source data, as discussed by Chim et al. [2025].

Study Objectives

This study was guided by four main objectives:

1. Examination of AI-in-education policies and guidance in Ireland, France, Australia, and the United States, alongside a review of evidence on public sentiment toward AI in education. Given the difficulty of collecting comprehensive and directly comparable sentiment data across countries, the study gave primary attention to European sentiment evidence, treating it as a cautious proxy for broader Western patterns where similar concerns, expectations, and debates are likely to emerge.
2. Assessment of selected NLP and related computational methods, including topic modelling, sentiment analysis, elements of causal NLP, and graph-based approaches, for linking policy language with stakeholder discourse. This supports the aim of examining how policy priorities, public perceptions, and implementation challenges relate to one another.
3. Generation and evaluation of synthetic data to address gaps, fragmentation, and limited comparability in available public sentiment datasets. Synthetic data was used to support a more balanced relationship between the policy and public sentiment corpora, with validation techniques used to assess the faithfulness of the synthesised data against the original evidence.
4. Development of a multi-layered analytical framework for analysing the policy–practice gap in AI-in-education governance, with particular attention to operational guidance, teacher preparedness, ethical use, and equity of access.

The dissertation was organised as follows. Chapter 2 reviews the literature on AI governance, public sentiment, and computational approaches to policy analysis, and

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identifies the gaps that motivate the study. Chapter 3 sets out the research methodology, including the rationale for the selected analytical methods, experiments, and evaluation criteria. Chapter 4 describes the implementation of the NLP techniques and presents the experimental results. Chapter 5 critically interprets these results in relation to the research question and the wider literature. Finally, Chapter 6 summarises the main findings, discusses the study's limitations, and outlines directions for future research.

Chapter 2

Literary Review

This chapter reviews the literature that informs the study of the policy–practice gap in AI-in-education governance. It focuses on how AI policy is framed, how its implementation is discussed, and how public sentiment helps reveal the conditions under which AI adoption is accepted, questioned, or resisted in educational settings.

The review is organised around three connected areas. Section 2.1 examines national policy and guidance documents from Ireland, France, Australia, and the United States in order to identify how different governance systems define AI-related priorities, risks, responsibilities, and expectations for implementation. Section 2.2 reviews evidence on public sentiment and stakeholder perspectives, with particular attention to European evidence as a cautious proxy for wider Western trends. This part of the review considers how teachers, students, parents, and the wider public perceive issues such as trust, fairness, ethical use, teacher preparedness, access, and the social consequences of AI in education. Section 2.3 then reviews computational approaches to policy and sentiment analysis, especially NLP methods such as topic modelling, sentiment analysis, and selected elements of causal NLP, as well as the use of synthetic data to address gaps and imbalance in available sentiment data.

By bringing these areas together, the chapter establishes the conceptual and methodological basis for the study. It shows that AI governance, public sentiment, and educational implementation are often examined separately, even though they are closely connected in practice. The chapter therefore identifies the need for a multi-layered, data-driven framework capable of linking policy language, stakeholder discourse, sentiment evidence, and implementation challenges in the analysis of AI-in-education governance.

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2.1 National Policy Frameworks

As outlined in Chapter 1, Ireland, France, Australia, and the United States were selected to provide a varied set of governance contexts for examining AI education policy. This section develops that comparison by reviewing each national framework in turn, with attention to how policy priorities, regulatory assumptions, implementation responsibilities, and educational risks are framed.

The purpose of this review is not only to describe national policy positions, but to identify how different governance settings shape the relationship between policy intention and educational practice. The country-specific subsections that follow therefore examine each case in relation to the wider policy–practice gap, focusing on the extent to which guidance is made operational, responsibilities are clarified, and implementation challenges are addressed.

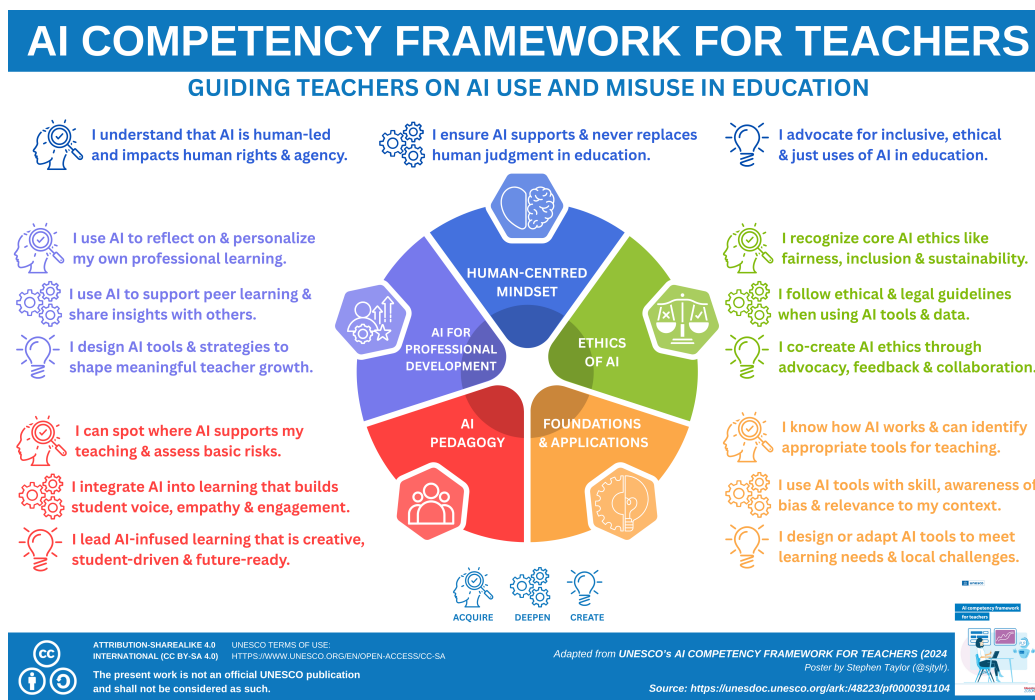


Figure 2.1: Overview of UNESCO’s *AI Competency Framework for Teachers*, whose emphasis on teacher capability, ethical awareness, pedagogical use, and professional learning is reflected in Ireland’s emerging AI-in-education policy and guidance. Source: adapted from the framework by Miao and Cukurova [2024].

2.1.1 Ireland

Ireland’s approach to AI in education combines a high-level digital strategy with emerging, sector-specific guidance responding to the rapid development of GenAI. According to Department of Education [2022], the national agenda is framed by the *Digital Strategy for Schools to 2027*, which aims to support digital transformation and strengthen digital competence among teachers and learners while aligning Irish policy with wider European digital education priorities. As discussed in Chapter 1, this direction is broadly consistent with the UNESCO *AI Competency Framework for Teachers* by Miao and Cukurova [2024], which emphasises that AI integration depends not only on access to technology but also on teacher capability, ethical awareness, pedagogical use and professional learning. Figure 2.1 provides an overview of the framework’s main dimensions, covering a human-centred mindset, AI ethics, foundations and applications, pedagogy and professional development. The 2024 Implementation Plan by Department of Education [2024] makes this agenda more operational by setting actions for teacher professional learning, digital infrastructure, online safety, digital leadership and AI-specific supports. It records important completed actions, including an AI Hub for teachers and school leaders, digital leadership training, digital communities of practice for DEIS schools supported through Ireland’s *Delivering Equality of Opportunity in Schools* programme, and online safety resources. At the same time, further work remains in progress on professional learning, infrastructure guidance, procurement frameworks, cybersecurity, EU engagement and AI-specific support for schools and classroom routines.

As set out by Department of Education and Youth [2025], the *Guidance on Artificial Intelligence in Schools* builds on this wider strategy by offering more direct support to teachers and school leaders. It sets out principles for safe, ethical, and appropriate AI use, with attention to privacy, data protection, human oversight, transparency, bias, and the need for teachers to verify AI-generated outputs. It also points schools towards practical supports, including the AI in Schools Hub, an introductory online course, and a 4P approach based on purpose, planning, policies, and classroom practice.

However, the guidance also illustrates the limits of the current policy response. It is presented as a first version and a living document, which suggests that many aspects of implementation remain unsettled, especially assessment, monitoring, teacher

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readiness, and the evaluation of AI’s impact on learning. As set out by Smeaton et al. [2025], the AI Advisory Council’s *AI and Education Advice Paper* makes this point more explicitly, arguing that the rapid uptake of GenAI has created uncertainty for teachers, students, and parents around appropriate use, assessment, and academic integrity. The Council also warns that AI detection tools are not a reliable solution, and calls instead for constructive use, stronger AI literacy, clearer guidance, and regular policy updates. Its equity concerns are also significant, particularly in relation to unequal access to paid AI tools and the limited support for the Irish language, which may disadvantage Gaelscoileanna.

In response to these uncertainties, local institutions and education bodies have developed their own policies, focusing on ethical use, academic integrity, and practical guidance for learners, as shown by The Ark HQ [2025] and Cavan and Monaghan Education and Training Board [2023]. This local response points to the limited availability of fully practical national guidance and shows how schools are interpreting policy within their own contexts. At a broader level, Organisation for Economic Co-operation and Development (OECD) [2025] similarly stresses the need to evaluate how digital technologies affect student learning, strengthen digital competence, and support equitable access, drawing on peer experience from systems such as Estonia, France and Sweden.

Research from Irish institutions reinforces this mixed picture. Byrne et al. [2024] note that GenAI may support lesson preparation, personalised learning, feedback, and administrative efficiency, while raising concerns about reliability, data privacy, unequal access, academic integrity and responsible use. They further note that the high cost of developing foundation models concentrates capacity among a few providers, leaving many institutions dependent on external tools. Ongoing policy development reflects this same tension. The *First Interim Report*, presented by Joint Committee on Artificial Intelligence [2025], calls for a coordinated national approach to AI governance, stronger public and institutional AI literacy, effective implementation of the EU AI Act discussed in European Parliament and Council of the European Union [2024], and safeguards for transparency, equality, children’s rights and equitable access. Taken together, these developments illustrate a system in transition: strategic ambition is clear, but implementation and oversight remain uneven, leaving a visible policy–practice gap in guidance, teacher support and institutional responses to AI in education.

2.1.2 France

France’s approach to AI is shaped by a strong national ambition to reduce technological dependence and position the country as a leading European actor in AI. The report *Our AI: Our Ambition for France* by Artificial Intelligence Commission [2024] presents AI as both an economic opportunity and a sovereignty challenge. It calls for investment in computing capacity, better data access, research systems and training, alongside public-sector transformation that remains human-centred, responsible and internationally governed. Within education, this ambition provides the wider context for a more structured approach to AI literacy, teacher support and regulation of AI use in schools.

This strategic ambition is accompanied by a regulatory approach shaped by both European and national frameworks. As described by NewMind AI [2025], France’s AI policy is closely linked to the EU AI Act, while also drawing on domestic measures in areas such as data protection, intellectual property, algorithmic transparency, and sector-specific oversight. Rather than developing a single comprehensive national AI law, France appears to be adapting existing legal and policy instruments to AI-related challenges. The French case therefore illustrates a relatively coherent model in which EU regulation provides the main legal structure, while national measures translate these principles into domestic governance practice.

At the level of educational practice, Ministère de l’Éducation Nationale [2025] introduced *L’IA en éducation : cadre d’usage*, a framework for the ethical, legal, and pedagogical use of AI in schools. It authorises AI use only within defined conditions, distinguishes between predictive and GenAI, warns against non-sovereign public tools where personal or protected data are involved, and stresses transparency, critical verification, and the adaptation of assessment practices. This approach also aligns with UNESCO guidance on GenAI in education by Giannini [2023], which stresses critical, ethical, and pedagogical teacher preparedness. National research from the *Direction du Numérique pour l’Éducation*, presented in *Direction du Numérique pour l’Éducation* [2024], further supports this orientation by reviewing AI in education through public policy, ethics, classroom applications, teacher training, and recent developments in GenAI. The report highlights the need to develop critical thinking, data awareness, ethical judgement and a basic understanding of AI systems, while ensuring that AI supports rather than replaces teachers’ professional

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role. Together, these elements suggest a deliberate effort to embed AI in education through teacher development, ethical guidance and structured pedagogical use. They also link AI literacy with assessment, data protection, and teacher agency.

Despite this structured policy environment, implementation challenges remain. The report by Montreuil et al. [2025] states that AI use in schools is still uneven and often shaped by individual teachers or local initiatives rather than a shared institutional strategy. It identifies teacher training as a major weakness, calling for clearer guidance, practical support and a large-scale national training effort. The report also stresses stronger coordination across school, academic and national levels to support consistent and equitable deployment in practice over time. Empirical evidence from the AI4T project gives a more practical view of these training needs and classroom adoption barriers. The French national evaluation by Paris et al. [2023] shows that targeted professional learning can strengthen teachers’ understanding of AI and support more informed classroom use. However, it also suggests that participation and impact depend on access to training, institutional support, teacher confidence and time for experimentation.

Overall, the French case shows a system that is comparatively well-structured at the policy level, but still constrained by uneven implementation in practice. Its main challenge is to translate national ambition and regulatory clarity into consistent classroom-level capacity, while balancing pedagogical innovation with digital citizenship, student development, and technological sovereignty.

2.1.3 Australia

The rapid rise of GenAI in late 2022 led to a swift and coordinated response across Australian education. At the national level, this response is set out in the *Australian Framework for Generative Artificial Intelligence in Schools* by Department of Education [2023], developed with input from jurisdictions, school sectors and national education agencies. The framework offers guidance for policymakers, school leaders, teachers, support staff, parents and students. Its principle-based design fits Australia’s federal school system, where responsibilities are shared across states, territories and sectors in practice and policy development. Its six principles cover teaching and learning, wellbeing, openness, fairness, accountability, privacy, security, safety, and responsible implementation.

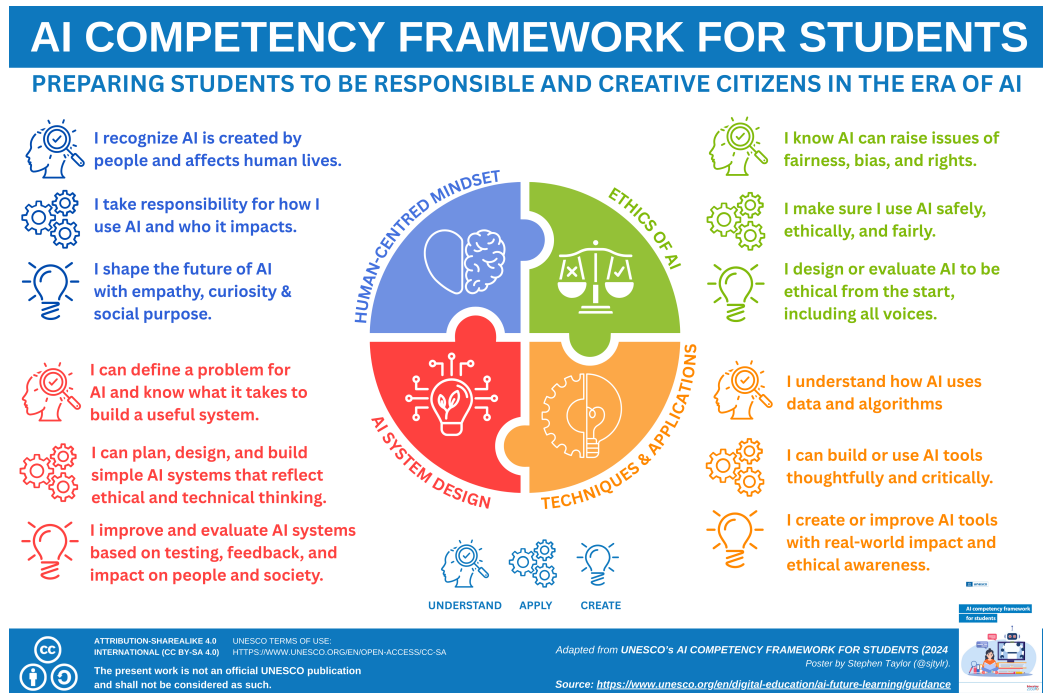


Figure 2.2: Overview of UNESCO's *AI Competency Framework for Students*, which complements the teacher framework by focusing on human-centred AI, ethics, AI techniques and applications, and AI system design for responsible and creative student participation in the AI era. Source: adapted from the framework by Miao et al. [2024].

Complementing the national framework, Brazil et al. [2025] present the PATH principles as a practical model for AI use in teaching and learning. The model draws on international guidance, including UNESCO's AI competency frameworks for teachers and students by Miao and Cukurova [2024] and Miao et al. [2024]. Figure 2.2 gives an overview of the student framework, which focuses on responsible AI use, ethical awareness, basic technical knowledge, and AI system design. The PATH principles, namely Promote teaching and learning, Advance wellbeing and social interaction, Train for AI literacy, and Harness AI ethically, translate these concerns into practical priorities for both teachers and learners. They stress that AI should support teaching and learning without replacing teacher judgement or student agency, while protecting wellbeing, sustaining social interaction and building AI literacy within commitments to responsibility, privacy, fairness and human rights. Read alongside the Australian national framework, the PATH model links policy intent with classroom practice through a human-centred approach to AI use.

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Despite this relatively clear national direction, a gap remains between policy principles and day-to-day implementation. Reports by Ingvarson and Havinga [2023] and Knight et al. [2023] describe a pragmatism gap, where schools must turn broad ethical principles into practical action within a fast-moving and commercially complex AI ecosystem. This helps explain why institutions such as The Hutchins School and Virtual School Victoria have developed local policies that combine privacy, academic integrity and teacher judgement with requirements for transparency, consent, equity and tool review before use, as shown by The Hutchins School [2025] and Virtual School Victoria [2025]. Recent Australian research by Lodge and Loble [2023] adds that implementation must consider AI’s cognitive effects, especially harmful cognitive offloading when students use AI to avoid the effort required for knowledge building and critical thinking in classroom practice contexts.

Efforts to address these challenges are reflected in local and national developments. Although the 4E (Embrace, Enable, Experiment, and Exploit) framework by Shailendra et al. [2024] was proposed for university curriculum adoption, its focus on stakeholder roles, curriculum redesign, enabling staff and students, and evaluation offers a reference for school-level implementation. At school level, the 2024 review of the Australian framework by National AI in Schools Taskforce [2025] confirms continuing national work on workload, data security and privacy, equity, market power, research and copyright. Overall, Australia presents an early and well-framed response, whose impact depends on consistent translation into classroom practice.

2.1.4 United States

The United States’ approach to AI in education is characterised by a decentralised but rapidly evolving policy landscape, where federal guidance sits alongside strong state and local autonomy. At the federal level, the U.S. Department of Education provides a broad ethical and policy framing for AI in teaching and learning. As set out by U.S. Department of Education, Office of Educational Technology [2023], it highlights possible benefits such as teacher support, stronger adaptivity and responses to learning gaps, while warning about bias, privacy, surveillance, inequity and the scaling of automated decisions in education. However, implementation remains largely the responsibility of states, districts and institutions, creating a varied and sometimes fragmented governance landscape.

State-level initiatives play a central role in turning federal principles into practical guidance. Washington State, for example, promotes a “Human-AI-Human” (H-AI-H) framework, where AI use begins with human inquiry and ends with human reflection, insight and judgement, as set out by Washington Office of Superintendent of Public Instruction [2024]. Similarly, Massachusetts Department of Elementary and Secondary Education [2025] provides detailed guidance that connects ethical use, equity and legal compliance with AI literacy and teaching both with and about AI. Other states, including Wyoming and Oregon, offer more operational guidance that links policy development and privacy with governance, academic integrity, and responsible use, as shown by Wyoming Department of Education [2025] and Oregon Department of Education [2023]. Together, these examples show how states are translating broad AI principles into local implementations. However, Panorama Education [2024] still suggests an uneven landscape, where educators and district leaders continue to seek clearer and more consistent national guidance.

At the institutional level, implementation varies across districts and educational settings. Reich et al. [2025] describe the current phase as one of iterative policy development, where schools are effectively “building the plane while flying it,” favouring flexible, experimental approaches over fixed regulatory models. Some districts gradually expand student autonomy under teacher oversight, while others focus on teacher support and workload reduction, as shown by Northport-East Northport Union Free School District [2025] and Excelsior Multi-Academy Trust [2025].

Despite these efforts, significant concerns remain about equity and the wider social implications of AI use in education. Pennsylvania Advisory Committee [2024] highlights risks related to surveillance, bias, unequal access, and the possible weakening of teacher-student relationships if AI systems are introduced without strong safeguards. This creates a tension between innovation-driven initiatives and the need to protect civil rights, student privacy, and vulnerable populations. The same concern appears in state-level toolkits and recent academic work, which stress the importance of transparency, human oversight, teacher support, and equitable implementation, as shown by Oklahoma State Department of Education [2025] and Medina [2026]. Overall, the U.S. case shows how decentralisation can encourage experimentation and rapid policy development, but also contributes to fragmentation and uneven implementation across the education system, which in turn raises concerns about equity, oversight, and the social consequences of AI in education.

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The preceding sections show how national and institutional policies define the conditions for AI adoption in education. Yet policy documents alone cannot explain how AI is understood, trusted, questioned or resisted by those affected by its use. For this reason, the review now turns to public sentiment. This perspective connects formal governance with lived educational experience, while showing where policy ambition may diverge from stakeholder concerns, expectations and everyday practice in schools and classrooms.

2.2 European Public Sentiment

As mentioned in Chapter 1, the public sentiment component of this study is narrower than the policy review. The policy analysis includes Ireland, France, Australia and the United States, because looking beyond Europe can show whether faster-moving cases offer lessons for gaps visible in European policy and sentiment data. By contrast, directly comparable public sentiment data remains harder to obtain across all these contexts. This section therefore focuses mainly on European data on attitudes toward AI in education. This focus is supported by convergence in European research, where stakeholders often show cautious optimism about efficiency, personalised learning and innovation while raising concerns about privacy, bias, transparency, academic integrity, human agency and the changing role of teachers in classroom practice.

The section first reviews European-level evidence on AI and education, drawing on studies that capture the views of young people, educators, students, school leaders, and early adopters of GenAI in secondary education. It then briefly considers public sentiment in Ireland and France, since these are the two European countries targeted in this study. This structure connects three layers of evidence: European sentiment data, national policy and public perception cases, and external policy examples that may help interpret European policy–practice gaps. The evidence reviewed here also shows why additional methodological support is needed later in the study. Compared with the relatively rich set of policy documents available, sentiment material is more uneven and less balanced across stakeholder groups and national contexts. For this reason, these sources, together with additional documents, provide the empirical basis for constructing an augmented textual corpus that can support a more balanced sentiment and policy–practice analysis.

2.2.1 European-level evidence

European-level evidence suggests that public sentiment toward AI in education is marked by cautious and conditional acceptance. Stakeholders generally recognise that AI, especially GenAI, may support learning and feedback while reducing routine workload and creating new opportunities for inclusion and communication. This pattern is also visible in a recent survey by European School Education Platform Editorial Team [2026], where respondents reported using AI mainly for lesson planning and teaching materials, as well as assessment preparation, explanations, practice exercises and administrative tasks. However, these benefits are balanced against concerns about privacy, data use and bias, alongside misinformation, academic integrity, pupil over-reliance, unclear guidance and the risk that automated tools may weaken human judgement or educational relationships, as shown by Stefan [2024] and Villar-Onrubia et al. [2025].

Research on young people brings in a rights-based perspective. As shown by Stefan [2024], young people are not only learners or users of AI systems, but also data subjects and citizens affected by algorithmic profiling, platform governance, and automated decision-making. From this perspective, AI literacy includes not only technical skills but also an understanding of how AI shapes participation, rights, well-being and democratic agency. Evidence from secondary education shows that students and educators are already experimenting with GenAI, often where guidance, training and support remain insufficient. Early adopters report value in lesson preparation, feedback, summarisation and student support, while stressing the need for teacher development, approved safe AI tools, clearer rules and stronger ethical safeguards, as shown by Villar-Onrubia et al. [2025] and European School Education Platform Editorial Team [2026].

Overall, European data points to a policy–sentiment gap not in the stated values of policy, but in their operational translation. Stakeholders often echo policy goals around human-centred, transparent and accountable AI, yet survey results reported by European School Education Platform Editorial Team [2026] show that AI is already entering school practice before safe tools, assessment rules, training and clear local guidance are fully in place. This makes public sentiment a useful complement to policies, because it shows where formal guidance remains too general for the practical concerns experienced by teachers, students, youth workers and school leaders.

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2.2.2 National Sentiment Patterns in Ireland and France

In Ireland, public sentiment around AI reflects the wider European pattern of cautious optimism, where interest in innovation is tempered by concerns about trust, regulation, transparency and long-term social impact. Irfan et al. [2023] and Kirk et al. [2023] suggest that Irish society recognises AI’s potential across education, healthcare and industry, but acceptance remains conditional on accountability and effective governance. These concerns are also visible in schools: students value AI for efficiency and personalised learning but worry about critical thinking and academic integrity, while educators recognise similar opportunities yet report limited training, unclear guidance and uneven resources, as shown by Butler et al. [2023]. Large-scale evidence by Quality and Qualifications Ireland [2025] shows widespread GenAI use alongside uncertainty around appropriate use, assessment and policy clarity. Youth perspectives add a rights-based dimension, with young people seeing AI as both an opportunity and a risk for fairness, safety and participation in decision-making, as shown by Ombudsman for Children’s Office [2024]. Together, these findings make the European policy–sentiment gap visible in Ireland: AI is promising, but educational use is limited by weak guidance, uneven inclusion and support gaps.

In France, public and stakeholder sentiment around AI is visible mainly through national surveys, European evidence, and research on digital technology use, but it remains only partly focused on GenAI in education. Much of the evidence concerns broader AI or digital adoption rather than classroom practice, which limits its precision. Even so, the available sources point to a consistent pattern: AI use is growing, but trust, guidance, and equity remain central concerns. In practice, AI adoption among students is rising quickly. European-level data from Vodafone Foundation [2025] suggests that nearly half of French students aged 12–17 use tools such as ChatGPT, both independently and in classroom contexts, showing growing familiarity with AI-supported learning. Students also recognise the importance of AI skills for future careers, although regional inequalities and the digital divide remain relevant. At societal level, Agence Nationale de la Cohésion des Territoires [2025] and Brice Mansencal et al. [2025] show increasing use of AI tools alongside caution about their role in work and everyday life. Overall, the French case reflects a familiar European pattern: adoption is moving faster than institutional guidance, teacher support, ethical safeguards, and equitable access to AI competencies.

2.3 NLP for Policy and Sentiment Analysis

The preceding sections reviewed the policy frameworks and public sentiment evidence that shape AI in education. Together, they show that the policy–practice gap is not located in policy documents alone, nor only in public attitudes, but in the relationship between governance language, stakeholder concerns, and the conditions of implementation. This section therefore turns to NLP-based and related knowledge-based approaches that support the analysis of these relationships across policy documents, guidance, public discourse, and stakeholder sentiment.

The section reviews six connected areas relevant to this study. First, it considers classical text-as-data approaches, including preprocessing, document representation, topic modelling, and methods for identifying recurring themes in large text corpora. Second, it examines language-model-based approaches for classification, annotation, summarisation, and discourse analysis. Third, it reviews sentiment analysis as a distinct layer for examining how teachers, students, parents, and the wider public express trust, concern, acceptance, or resistance toward AI use in education. Fourth, it considers causal NLP for exploring relationships between policy language, stakeholder sentiment, and implementation concerns. Fifth, it considers synthetic data generation as a response to gaps, imbalance, and limited comparability in available sentiment evidence. Finally, it briefly discusses knowledge-based techniques, including ontologies, knowledge graphs, and GraphRAG, before explaining why they remain outside the core scope of this study. In doing so, the section establishes the link between the literature review and the research design developed in Chapter 3.

2.3.1 Classical Text-as-Data Approaches

Having reviewed the policy frameworks and public sentiment evidence that shape AI in education, the review now turns to methodological approaches for analysing policy texts, stakeholder discourse, and public sentiment more systematically. Wider AI governance research shows that ethical principles and high-level frameworks often remain difficult to translate into operational processes, institutional responsibilities, and measurable forms of implementation, as shown by Birkstedt et al. [2023]. In the context of AI in education, this challenge is especially important because relevant evidence is distributed across policy documents, institutional guidance, stakeholder discourse, public commentary, and survey-based sentiment data.

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Text-as-data methods provide a way to examine these sources at scale while retaining a link with interpretation. Work by Grimmer and Stewart [2013] argues that automated text analysis can make large-scale political and policy text collections systematically analysable, while also stressing the need for problem-specific validation and close interpretive judgement. Similarly, Gentzkow et al. [2019] present text as a major source of empirical evidence for social science research, showing how statistical and computational methods can extract structured information from unstructured language data. Within policy research, Wilkerson and Casas [2017] show that large-scale computerised text analysis has been used to study political communication, policy agendas, public opinion, and institutional discourse.

Topic modelling is particularly important in this tradition because it provides a way to identify latent themes across large corpora. The original Latent Dirichlet Allocation model by Blei et al. [2003] remains a key reference for probabilistic topic modelling, while structural topic models by Roberts et al. [2014] and Roberts et al. [2016] extend this approach by incorporating document-level covariates, making them especially useful for comparing texts across institutions, time periods, or policy contexts. In policy studies, Isoaho et al. [2021] position topic modelling as a bridge between computational analysis and qualitative interpretation, helping researchers identify recurring themes and discursive patterns without replacing close reading.

Preprocessing techniques

Classical text-as-data methods depend on preprocessing because they usually represent documents through tokens, frequencies, dictionaries, or document-term matrices. Preprocessing therefore shapes what the model can detect and how results are interpreted. Standard steps include removing duplicates, cleaning formatting noise, standardising case, tokenising text, removing stop words, handling punctuation and numbers, and deciding whether to apply stemming, lemmatisation, or phrase detection, as outlined by Welbers et al. [2017]. These choices are not purely technical: they can affect whether institutional terms, stakeholder categories, negations, or domain-specific expressions remain visible in the corpus. Denny and Spirling [2018] show that preprocessing decisions can substantially affect unsupervised text-analysis results and therefore need to be tested rather than treated as neutral defaults. Stop-word removal is particularly consequential, as Schofield et al. [2017] show that it can change the interpretability and coherence of topic-model outputs.

The importance of preprocessing is especially clear in policy and education corpora, where sources often differ in length, formality, structure, and genre. Policy documents may contain headings, legal phrases, bullet points, references, and repeated institutional language, while public comments and survey responses may contain informal wording, spelling variation, emojis, abbreviations, or short expressions of approval and concern. In comparative and policy-oriented text analysis, Lucas et al. [2015] show that document preparation, translation, metadata, and source comparability can shape downstream interpretation. For this reason, preprocessing should be aligned with the analytical task and corpus characteristics. Topic modelling may require careful removal of boilerplate text and overly frequent institutional terms, while sentiment analysis requires caution because removing negation words or intensifiers can distort the meaning of evaluative statements. In this study, preprocessing is therefore treated as part of the research design rather than a neutral technical step.

2.3.2 Language Models for Discourse Analysis

Recent advances in language modelling have expanded the methods available for analysing policy texts, stakeholder discourse, and public sentiment. Earlier text-as-data methods, such as dictionary-based analysis, classical classifiers, and topic models, remain useful, but they often struggle with context, ambiguity, negation, and domain-specific meaning. Smaller transformer-based models, including BERT, RoBERTa, and DistilBERT, address some of these limits by producing contextual representations of text, as shown by Devlin et al. [2019], Liu et al. [2019] and Sanh et al. [2019]. This makes them useful for sentiment analysis, stance detection, document classification, and policy-topic classification, especially where views on AI in education are mixed or conditional. Related embedding-based approaches have also shaped topic discovery, including BERTopic, which combines transformer embeddings with clustering and class-based TF-IDF to generate interpretable topic representations, as proposed by Grootendorst [2022]. These methods are relevant to AI-in-education governance because they support more flexible analysis of heterogeneous sources, including policy texts, consultation responses, survey answers, and online discourse. They also help link broad patterns in stakeholder discourse with specific governance, implementation, and institutional concerns.

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Small Language Models (SLMs) and Large Language Models (LLMs) add a methodological layer. Recent work by Lu et al. [2024] and Nguyen et al. [2025] presents SLMs as compact models designed to provide useful language capabilities with lower computational cost, faster inference, and easier adaptation to specific tasks. This makes them relevant for document classification, sentiment detection, weak labelling, and policy-topic analysis, especially when the aim is to process large corpora in a more controlled and resource-efficient way. By contrast, recent social science research by Törnberg [2025], Heseltine and Clemm von Hohenberg [2024] and Törnberg [2024] has explored LLMs for annotation and classification, as well as for summarisation, qualitative coding, thematic analysis, and the generation of synthetic or weakly labelled data. These capacities are valuable when sources are heterogeneous, difficult to label manually, or unevenly distributed across document types. In policy and education research, SLMs and LLMs can play complementary roles. SLMs are useful when categories are defined and the task requires scalable, repeatable classification across policy documents, survey answers, or public comments. LLMs are better suited to exploratory analysis, including theme identification, code-book refinement, initial labelling, summarisation, and stakeholder comparison. This distinction is valuable for AI-in-education governance because evidence is distributed across policy texts, institutional guidance, survey evidence, and online discourse.

However, the use of language models also raises methodological concerns. SLMs depend on suitable training or adaptation data, task-specific labels, and validation against human-coded examples, while Weber and Reichardt [2024] and Törnberg [2024] show that LLM-based annotation requires careful evaluation because performance can vary across models, prompts, and annotation tasks. Recent studies by Gilardi et al. [2023], Heseltine and Clemm von Hohenberg [2024] and Ziems et al. [2024] show that LLMs can perform well on text-annotation tasks, including relevance, stance, topic, frame, sentiment, and political text classification, but they also emphasise that model outputs should be compared with human judgement rather than accepted uncritically. LLMs may also reproduce biases in training data or generate fluent interpretations that appear credible without being sufficiently grounded in the source text, making transparent prompting, source checking, and careful reporting of model choices essential, as discussed by Bender et al. [2021] and Törnberg [2024]. In policy and sentiment analysis, language models should therefore be treated as assistive analytical tools rather than independent sources of evidence.

2.3.3 Sentiment Analysis and Stakeholder Attitudes

Sentiment analysis provides a complementary perspective by focusing on the affective and evaluative dimensions of text. Classic work by Pang and Lee [2008] and Liu [2012] established the field as a way to identify polarity, subjectivity, and attitudes in textual data. More recent reviews by Tan et al. [2023] and Mao and Liu [2024] show that sentiment analysis now includes lexicon-based, machine-learning, deep-learning, and hybrid approaches, each with different implications for preprocessing, feature extraction, classification, and domain adaptation. For policy and education research, this is important because stakeholder discourse rarely expresses simple approval or rejection. Teachers, students, parents, and the wider public may support AI-assisted learning while also expressing concern about privacy, academic integrity, assessment, fairness, teacher autonomy, or unequal access.

In this context, sentiment analysis should be used cautiously and in combination with thematic interpretation. General polarity labels such as positive, negative, and neutral may be too coarse to explain how stakeholders evaluate AI use in education. For example, a comment may be positive about administrative efficiency but negative about assessment integrity, or supportive of personalised learning while expressing concern about data protection. This makes aspect-based and context-sensitive sentiment analysis more appropriate than treating each document as having a single overall sentiment. Work by Hutto and Gilbert [2014] on social-media sentiment analysis also shows that informal language, irony, abbreviations, intensity markers, and platform-specific expressions can affect sentiment detection, which is relevant when analysing public comments and online discourse. In the present study, sentiment analysis is therefore used not as a standalone measure of public opinion, but as a way to identify patterns of trust, concern, uncertainty, and resistance that can be compared with policy priorities and implementation challenges.

Education-focused studies illustrate this value. Recent work on GenAI in education uses sentiment analysis alongside topic modelling to examine how public discourse frames AI adoption, including perceived benefits, emotional reactions, and concerns about educational consequences, as shown by So et al. [2024], Liao et al. [2025], and Göçen et al. [2024]. These studies show that sentiment analysis is most useful when it is linked to the themes, actors, and contexts being discussed, rather than reported only as aggregate polarity. For this dissertation, the right use of sen-

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timent analysis is therefore interpretive and comparative: it helps reveal whether policy language around responsible, ethical, and inclusive AI is aligned with the concerns expressed by stakeholders, and where tensions emerge between policy ambition and public or practitioner experience across different evidence sources, stakeholder groups, and national contexts.

2.3.4 Synthetic Data and Faithfulness Evaluation

The analysis of public sentiment across countries and stakeholder groups is limited by data scarcity, imbalance, and weak comparability. Available evidence often comes from surveys, institutional reports, public comments, and academic studies that use different sampling strategies, question formats, and levels of detail. This creates an uneven corpus in which some countries, stakeholder groups, or sentiment positions are better represented than others. In this context, synthetic data is used as a controlled augmentation strategy rather than as a substitute for empirical evidence. Its purpose is to increase the analytical coverage of under-represented categories, support robustness testing, and make comparison between policy discourse and public sentiment more stable. This use is consistent with Argyle et al. [2023], which shows that language models can approximate aggregate patterns of public opinion under carefully specified conditions, and with wider synthetic-data research by Snoke et al. [2018] and Stanley and Totty [2024], which distinguishes between the utility of synthetic data and its limits as a replacement for real observations.

For this study, synthetic sentiment texts are relevant because they help address uneven evidence while keeping the analysis grounded in the original corpus and the reviewed literature. The generated data is treated as a task-specific approximation of stakeholder attitudes toward AI use in education, not as new empirical testimony. It is therefore linked to documented themes such as trust, ethics, assessment, teacher preparedness, equity, privacy, and implementation. Duplicate or near-duplicate items are understood as texts with highly similar content after normalisation, while irrelevant items are those with only weak relation to the study themes. These decisions matter because they shape which forms of sentiment and stakeholder concern remain visible in the corpus. Synthetic data is therefore used only as a supplementary layer for exploratory and comparative analysis, with all findings interpreted against the original empirical evidence.

The literature suggests that the faithfulness of synthetic text should be assessed through a multi-level distance framework rather than a single similarity score. Lexical similarity can be examined using vocabulary statistics and Jensen–Shannon distance over word distributions, while semantic similarity can be assessed with sentence embeddings and cosine similarity using Sentence-BERT-style representations, as proposed by Reimers and Gurevych [2019]. Distributional alignment can also be examined in embedding space using Maximum Mean Discrepancy (MMD), a kernel-based two-sample measure introduced by Gretton et al. [2012], and Fréchet-style embedding distance, following the broader logic of distributional evaluation in generative modelling by Heusel et al. [2017]. A further option is the classifier two-sample test (C2ST) by Lopez-Paz and Oquab [2016], where a classifier is trained to distinguish real from synthetic texts; performance close to chance suggests stronger similarity, while higher accuracy indicates detectable synthetic artefacts. In this view, faithfulness is understood in terms of usefulness rather than identity: synthetic data is valuable only insofar as it preserves task-relevant sentiment patterns, thematic structure, and semantic meaning while making the limits of generated evidence explicit, following Snoke et al. [2018].

2.3.5 Causal NLP and Relational Analysis

Causal NLP builds on a wider causal inference tradition that distinguishes causal claims from simple associations. Foundational work by Rubin [1974] introduced the potential-outcomes framework for estimating treatment effects in randomised and non-randomised studies, while Pearl [2009] developed a structural causal modelling approach for reasoning about causal assumptions, interventions, and confounding. These foundations matter for text-based research because policy and sentiment corpora can reveal associations between language, attitudes, and implementation concerns, but they do not automatically establish causal effects. Work by Egami et al. [2022] adapts this problem to text-as-data research by showing how text can be used as a treatment, outcome, confounder, or discovered measure in causal inference. Their framework is especially relevant here because it stresses careful research design, validation of text-derived measures, and the separation between measurement and estimation before any causal interpretation is attempted from textual evidence and linked back to the research question.

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In AI-in-education governance, causal NLP is useful because the policy–practice gap is not only a matter of identifying themes or sentiment labels, but also of exploring how policy priorities, stakeholder concerns, and implementation barriers may relate. For example, policy emphasis on ethics, assessment, teacher readiness, or equity may correspond to particular forms of public concern, institutional uncertainty, or practical difficulty. Recent work by Ma et al. [2025], Liu et al. [2025], and Wan et al. [2025] suggests that language models can support causal relation extraction, causal reasoning, and causal discovery from text. However, this work also shows that these approaches remain sensitive to prompts, assumptions, data quality, and validation. For this reason, causal NLP is best treated in this dissertation as an exploratory and interpretive lens rather than as a basis for strong causal claims. Its strength is that it helps move beyond isolated topics and sentiment scores toward a more structured account of possible relationships between policy language, stakeholder sentiment, and implementation concerns. Its weakness is that the available evidence is observational, fragmented, and uneven across countries and stakeholder groups, which limits causal identification and makes expert interpretation essential.

2.3.6 Knowledge-Based Alternatives

Other computational approaches could also be used to analyse policy and education discourse, including ontologies, knowledge bases, knowledge graphs, and Graph Retrieval-Augmented Generation (GraphRAG). Ontologies provide explicit conceptual structures for representing entities, relations, and categories in a domain, while knowledge graphs organise information as interconnected nodes and edges that can support semantic search, reasoning, and explainability. Recent work by Meroño-Peñuela [2025] demonstrates how knowledge graphs can structure rules, responsibilities, datasets, and governance processes in more transparent and machine-readable ways. Similarly, GraphRAG combines graph-based representations with retrieval-augmented generation, allowing systems to retrieve not only semantically similar text but also relational information between concepts, actors, and documents, as discussed by Peng et al. [2025].

These approaches are promising for future work, especially where the aim is to build a structured knowledge base of AI policy concepts, map relationships between policy actors, or support question answering over policy documents. Tools and

frameworks such as Protégé, OWL, RDF, SPARQL, and ontology-engineering methods can support the formal modelling of domain concepts, relationships, constraints, and inference rules, as shown by Musen [2015] and Horridge [2011]. However, they are not used as core methods in this study because the research question focuses on analysing textual patterns, sentiment, and implementation concerns rather than constructing a formal semantic representation of the policy domain. Ontology and knowledge-graph construction would also require additional design choices, such as schema definition, competency questions, entity linking, relation extraction, graph validation, and query evaluation, which would substantially expand the scope of the dissertation. For this reason, the study prioritises NLP methods that are better aligned with its corpus-based objective: identifying themes, sentiment patterns, and links between policy language and stakeholder concerns.

Summary

This chapter reviews the literature informing the study of the policy–practice gap in AI-in-education governance. It examines how AI policy is framed, how implementation responsibilities are discussed, and how public sentiment reveals the conditions under which AI adoption is accepted, questioned, or resisted in educational settings. The review treats this gap as a layered governance issue emerging between national policy ambitions, institutional enactment, stakeholder concerns, and the practical realities of schools and classrooms.

The chapter is organised around three connected areas. Section 2.1 examines national policy and guidance documents from Ireland, France, Australia, and the United States, comparing how different governance systems define AI-related priorities, risks, responsibilities, and implementation expectations. Section 2.2 reviews public sentiment and stakeholder perspectives, with particular attention to European evidence as a cautious proxy for wider Western trends. Section 2.3 reviews computational approaches to policy and sentiment analysis, including classical text-as-data methods, language models, sentiment analysis, synthetic data, causal NLP, and knowledge-based techniques, as summarised in Table 2.1. Together, these areas establish the conceptual and methodological basis for a multi-layered, data-driven framework linking policy language, stakeholder discourse, sentiment patterns, and implementation challenges.

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Table 2.1: Summary of NLP and related computational approaches in Chapter 2.3.

Approach	Main purpose	Relevance to this study	Key references
Classical text-as-data methods	Represent documents through tokens, frequencies, dictionaries, document-term matrices, topic models, and other structured features.	Provide a scalable way to examine policy documents, institutional guidance, and stakeholder discourse by identifying themes, dominant terms, discursive patterns, and differences across countries or stakeholder groups.	Blei et al. [2003]; Gentzkow et al. [2019]; Grimmer and Stewart [2013]; Isoaho et al. [2021]; Wilkerson and Casas [2017]
Preprocessing and corpus preparation	Clean, normalise, tokenise, and structure textual data before modelling, including decisions about stop words, stemming, lemmatisation, duplicates, boilerplate text, and domain-specific terms.	Important because preprocessing choices shape what remains visible in the corpus, including institutional concepts, stakeholder categories, negations, intensifiers, and AI-in-education terminology.	Denny and Spirling [2018]; Lucas et al. [2015]; Schofield et al. [2017]; Welbers et al. [2017]
Language models	Use contextual representations, SLMs, and LLMs for classification, annotation, summarisation, weak labelling, topic discovery, and discourse interpretation.	Support flexible analysis of heterogeneous sources such as policy texts, consultation material, survey answers, public comments, and online discourse. SLMs support repeatable classification, while LLMs support exploratory coding and interpretation.	Devlin et al. [2019]; Liu et al. [2019]; Lu et al. [2024]; Nguyen et al. [2025]; Törnberg [2025]; Ziemis et al. [2024]
Sentiment analysis	Identify affective and evaluative orientations in text, including polarity, subjectivity, trust, concern, uncertainty, acceptance, and resistance.	Helps compare stakeholder attitudes with policy priorities. It is most useful when combined with thematic interpretation, since views on AI use in education may be mixed, conditional, or aspect-specific.	Hutto and Gilbert [2014]; Liu [2012]; Mao and Liu [2024]; Pang and Lee [2008]; Tan et al. [2023]
Causal NLP and relational analysis	Use text as a treatment, outcome, confounder, or discovered measure to explore relationships between language, attitudes, and possible outcomes.	Useful as an exploratory lens for examining links between policy language, stakeholder sentiment, and implementation concerns, while avoiding strong causal claims from uneven observational evidence.	Egami et al. [2022]; Ma et al. [2025]; Pearl [2009]; Rubin [1974]; Wan et al. [2025]
Synthetic data and faithfulness evaluation	Generate supplementary textual data and assess similarity to original evidence using lexical, semantic, and distributional measures, including embedding distances and classifier-based tests.	Addresses data scarcity, imbalance, and weak comparability in public sentiment evidence, but requires validation of task-relevant sentiment patterns, themes, and semantic meaning.	Argyle et al. [2023]; Lopez-Paz and Oquab [2016]; Reimers and Gurevych [2019]; Snoko et al. [2018]; Stanley and Totty [2024]
Knowledge-based approaches	Use ontologies, knowledge bases, knowledge graphs, and GraphRAG to represent domain concepts, entities, relations, constraints, and links between documents or actors.	Relevant for future work, especially for structured knowledge bases or question answering over policy documents. They are not core methods here because the study focuses on corpus-based analysis rather than formal semantic modelling.	Horridge [2011]; Meroño-Peñuela [2025]; Musen [2015]; Peng et al. [2025]

Chapter 3

Methodology

3. METHODOLOGY

Chapter 4

Implementation and Results

4. IMPLEMENTATION AND RESULTS

Chapter 5

Analysis

5. ANALYSIS

Chapter 6

Conclusions

6. CONCLUSIONS

Chapter 7

Conclusion and Perspectives

7. CONCLUSION AND PERSPECTIVES

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